When Is a Rule a Function?

Algebra II Word Problems and Abstract Reasoning

Name: _____ Date: _____

Instructions

Suggested time: 45 minutes. Complete all 12 problems without a calculator. You may use this reference page. For every function decision, identify the input, the proposed output, and why the rule does or does not meet the definition. Support a rejection with a specific input or counterexample. Keep the stated domain and codomain unless a question asks you to change them.

The Definition

A function $f : D \to C$ assigns **exactly one element of C to every element of D.**

- **Domain D:** the full set of inputs the rule promises to accept.
- **Codomain C:** the declared set in which outputs must lie.
- **Range:** the outputs actually obtained, $\{f(x) : x \in D\} \subseteq C$.

Three Requirements

For each input in the stated domain:

1. An output must exist.
2. The output must be unique.
3. The output must belong to the stated codomain.

A rule can fail for different reasons at different inputs. Changing its domain, codomain, or assignment rule changes the proposed function.

What Is Allowed?

- Different inputs may have the same output. The function need not use every element of its codomain.
- Inputs and outputs need not be numbers. They may be names, ordered pairs, sets, or even functions. An entire set can be one output object.
- A function is **one-to-one** if different inputs always have different outputs. This is an additional property, not part of being a function.
- If the same input object can be represented in different ways, its output must not depend on which representation is used.

Relations and Composition

A relation from D to C is a collection of ordered pairs (x, y) with $x \in D$ and $y \in C$. It represents a function on D only if each element of D appears as a first coordinate with exactly one second coordinate.

For $(g \circ f)(x) = g(f(x))$ to be defined on all of D, f must be defined on D and every resulting $f(x)$ must be an allowed input to g.

When Is a Rule a Function? — Practice

1. A school has student IDs $S = \{A, B, C, D\}$ and lockers $L = \{1, 2, 3, 4, 5\}$. Its complete assignment record is

$$A \mapsto 2, \qquad B \mapsto 2, \qquad C \mapsto 4, \qquad D \mapsto 5.$$

   Does the record define a function from S to L? State its range and whether it is one-to-one. A second rule takes any locker in L and returns the ID of a student assigned to it. Is that a function from L to S? Identify every type of failure that occurs.

2. A library has a fixed inventory B of book IDs and a set S of registered students. At noon, one student has no borrowed books, another has two different books, and every loan is recorded. Compare these rules:

   (a) Input a student in S; output a book ID in B that the student has borrowed.

   (b) Input a student in S; output the complete set of that student's borrowed book IDs. The codomain is the collection of all subsets of B, including the empty set.

   Which rules are functions? Explain what happens for both described students.

3. Two weather stations each report at 8 a.m. and 8 p.m. on one fixed date. North reports 6° and 11°; South reports 4° and 4°. A rule takes a station name and returns a reported temperature from that date. Does this give a function from station names to real temperatures? Would using the ordered pair (station, reporting time) as the input repair the rule? State the full new domain and explain.

When Is a Rule a Function? — Continued

4. A courier accepts parcel masses $m \in (0, 5]$ kilograms. Its fee chart says "\$5 when $0 < m \leq 2$" and "\$8 when $2 \leq m \leq 5$." Does the chart define a function from accepted masses to dollar fees? Repair it by changing exactly one endpoint inequality while keeping the full domain. If both listed fees had instead been \$5, would the original overlap prevent the chart from defining a function? Explain.

5. A design program accepts a positive area A and returns the perimeter of a rectangle having that area, with no further restrictions. Does this define a function from positive real areas to positive real perimeters? Give two specific rectangles to support your answer. If the program is restricted to squares, does the rule become a function? Give its formula and its domain and codomain.

6. An integer machine accepts any integer n and must output an integer m satisfying $2m = n$. Does this define a function $\mathbb{Z} \to \mathbb{Z}$? Explain what happens for even and odd inputs. Give two separate repairs: one that changes only the domain and keeps infinitely many inputs, and one that changes only the codomain while keeping every integer input. State both repaired functions precisely.

When Is a Rule a Function? — Continued

7. A machine accepts a real number x and returns a real number y satisfying $y^2 = x$. Determine what happens when $x < 0$, $x = 0$, and $x > 0$. Does restricting inputs to $x \geq 0$ alone make the rule a function into $\mathbb{R}$? Give a domain and an output restriction that produce a function with infinitely many inputs, and write its rule.

8. A program takes a rational number r. It writes $r = p/q$, where p is an integer and q is a positive integer, and outputs $p + q$. Fractions need not be in lowest terms. Does this define a function $\mathbb{Q} \to \mathbb{Z}$? Use the input $r = 1/2$ to justify your answer. Would the assignment $p + q$ define a function if the input were instead the ordered pair (p, q)? State that new domain and explain the difference.

9. A program accepts any nonempty subset S of $\{1, 2, 3\}$. A set has no intrinsic order: $\{1, 2\} = \{2, 1\}$. For each proposed output below, decide whether the rule defines a function into $\{0, 1, 2, 3\}$. For each valid function, state its range.

   (a) The number of elements of S.

   (b) The first number written when someone lists the elements of S.

   (c) The smallest element of S.

When Is a Rule a Function? — Continued

10. A program's input is an entire real quadratic polynomial $p(x) = Ax^2 + Bx + C$, with $A \neq 0$. One version returns the real number $p(2)$. Does it define a function from quadratic polynomials to $\mathbb{R}$? Is it one-to-one? Exhibit two different inputs to support your answer. Another version returns a real zero of the input polynomial. Does that version define a function on the same domain? Give separate examples showing every type of failure that occurs.

11. Three students have advisor assignments $S_1 \mapsto A_1$, $S_2 \mapsto A_1$, and $S_3 \mapsto A_2$. The advisor directory contains A_1, A_2, and A_3. Advisors A_1 and A_2 both have room 101; A_3 has no assigned room. Consider the student-to-advisor rule and the advisor-to-room rule, using all listed students and all listed advisors as their respective domains. Is each rule a function? Does "look up a student's advisor, then that advisor's room" nevertheless define a function on all three students? Explain, and state a restriction of the advisor domain that makes the second rule a function and permits the composition.

12. Invent a real-world assignment with exactly four distinct input objects, a codomain of exactly four distinct output labels, and a range of exactly three labels. List all input–output pairs and explain why your rule is a function but is not one-to-one. Then alter the relation, keeping the same stated domain and codomain, so that it is no longer a function. Identify the precise requirement that your altered relation fails.